Physical Consistency Starts at Supervision: Evidential Actor Surface Fields for Dynamic Driving Worlds

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Abstract

Driving-world reconstruction can appear safer by removing uncertain Actors, including the legitimate hazards that matter for evaluation. We instead define physical consistency from ground-truth ray and surface supervision while keeping Actor identity, trajectory, extent, and hazard state immutable. Our compiler first separates image-rendering Gaussians from an Actor-canonical physical surface and exposes observed free, occupied, and unknown evidence without turning unknown mass into a deletion rule. On frozen Argoverse 2 data, the non-learned compiler reduces LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m while retaining every Actor and hazard state. We then train geometry on nuScenes: each completion seed generates a small set of child surfaces under motion-corrected target-set, local-plane, scale, per-frame coverage, literal first-return, and free-before-hit losses. With no deployment filter, this model reduces hazardous early returns by 5.12%, improves Chamfer by 6.21 mm, and raises hit recall by 2.76 points on an exposed development fold. Frozen producer evidence further predicts continuous FREE/OCCUPIED/UNKNOWN authority. Interpreting that authority as a categorical surface-return measure reduces hazardous early returns by 0.62 points and raises hazardous hit recall by 2.32 points relative to unit energy; treating it as additive opacity fails. Finally, we derive an exact boundary for normalized mixtures: attenuating component j lowers the pre-hit CDF iff its component CDF exceeds the current mixture CDF, and attenuation magnitude cannot reverse this sign. A failed learned-visibility ablation raises the boundary CDF on 59.32% of 99,208 rays, matching the predicted non-monotonicity. Typed ownership and inverse-query SE(3) composition keep appearance, canonical geometry, and rigid motion structurally separate. A single frozen 20-log nuScenes-to-AV2 evaluation is pending, so we claim source-domain supervised physical completion and a safety boundary, not domain invariance or real-road safety.

1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their visual geometry, however, is rarely a literal physical model. A Gaussian can explain image appearance while occupying observed free space, duplicating an Actor shell, or flickering across time. Conversely, a valid nearby Actor can be uncertain precisely because it is partially observed. Deleting that Actor improves an early-collision proxy without making the world more faithful. Physical consistency must therefore be defined before inference—through targets, supervision, representation, and state ownership—rather than certified after inference by a confidence filter.

We study this problem for dynamic annotated Actors. Multi-frame LiDAR is transformed into an Actor-canonical frame and split into build evidence and disjoint target sweeps. Actor identity, trajectory, extent, lifecycle, and hazard state remain immutable. Observed anchors are also fixed; each completion seed instead predicts a small set of child surfaces trained by target-set, local-plane, scale, per-frame coverage, literal first-return, and observed free-before-hit losses. Every child remains present at deployment. Thus early-return, hit-recall, and Chamfer improvements must come from the learned physical representation, not from UNKNOWN masking or Actor deletion.

This construction exposes a second distinction: surface geometry and sensor termination are different physical variables. A well-placed Gaussian can still receive contradictory free and occupied evidence from different views. We retain that evidence at the data producer and supervise continuous FREE/OCCUPIED/UNKNOWN authority for anchors and completion children. Treating the occupied coordinate as additive opacity fails because total optical thickness depends on primitive density. We instead compose frozen geometry and learned authority into one normalized categorical surface-return measure. Its CDF is the deployed first-return distribution, so training and inference share the same object and no post-hoc threshold is needed.

Dynamic motion, static context, and appearance are kept outside this geometry–termination coupling. Rigid Actor pose relocates the canonical field through inverse SE(3) querying but cannot deform it. Static Background has a different owner. Image gradients update only a sibling visual state through a stop-gradient physical carrier. These typed interfaces do not solve pose, visibility, or appearance; they prevent those variables from silently compensating for physical geometry during optimization.

Finally, the normalized return measure permits an exact safety analysis. Attenuating one primitive does not necessarily lower probability before a hazardous boundary: the sign depends on whether that primitive's own pre-boundary CDF lies above or below the mixture CDF. We derive the finite change and show that attenuation magnitude cannot reverse its sign. A failed learned- visibility experiment then becomes an interpretable boundary rather than motivation for another filter. We verify this effect on 99,208 frozen rays and never use the target-dependent sign as a deployment gate.

Our contributions are:

- supervision-native Actor completion whose GT set, ray, and free-space losses remain active through physical training, with no UNKNOWN or confidence deletion at deployment;
- producer-evidential FREE/OCCUPIED/UNKNOWN authority represented as a categorical surface-return measure rather than density-dependent volumetric opacity;
- typed canonical-geometry, rigid-pose, static-world, and appearance ownership with inverse-query SE(3) equivariance and no image-gradient path to physical geometry;
- an exact finite attenuation boundary for normalized mixtures and a frozen nuScenes-to-AV2 protocol that withholds cross-domain claims until its complete 20-log aggregate is available.

2. Related Work

Driving-world reconstruction. nuScenes [4] and Argoverse 2 [31] provide calibrated multi-sensor trajectories. LiDAR is 2.5D ray evidence: points discard free-space intervals [10]. Scene graphs model Actors [22], while 3D Gaussian Splatting models appearance [16]. UniSim targets closed-loop neural sensor rendering [35]; LiDAR4D and DyNFL target space–time LiDAR synthesis and re-simulation [32, 39]. These optimize observations, not collision semantics. We ask whether an edited Actor owns a queryable, first-return-consistent surface with explicit unknown space; Gaussian opacity is not occupancy, and our surface preserves provenance.

Neural LiDAR Fields models first returns through a physically motivated ray-weight distribution [11], while categorical depth supervision encourages a sharp metric depth mode [23]. QueryOcc instead samples positive and negative LiDAR queries directly in continuous space [19]. We combine their relevant boundary conditions: native GT rays define supervision, but an Actor-local 3D score is normalized only when answering a ray. This separates learned geometry evidence from the sensor termination distribution and keeps image, motion, and hazard variables outside the geometry head. Ray potentials constrain visibility [26], while evidential occupancy renders its first occupied hit back to LiDAR depth [14, 15]. We separate that literal hit from a target-nearest proxy and name the operator used by every audit. ALSO derives occupied and free supervision directly from the LiDAR sensor origin [3], while SCPNet shows that multi-frame completion labels can contain moving-object traces [33]. DualAD instead carries static and dynamic state in separate streams with explicit ego/object-motion compensation [6]. We use these observations to audit the producer of Actor-local targets and to supervise a continuous physical field; they do not justify deleting an inconsistent point after inference.

Recent 4D formulations make motion a first-class factor rather than a geometry residual. DynamicVGGT equips Gaussians with velocity under scene-flow supervision [9]; DeGO separates rigid offsets from non-rigid deformation [7]; and SelfOccFlow predicts separate static/dynamic signed-distance fields before flow-warped temporal aggregation [30]. These works motivate our separation boundary, not a visual-teacher shortcut: canonical surface, visibility, and temporal deformation must receive distinct physical supervision.

Learned geometry completion. SelfOcc treats the reconstructed volume as a continuous signed-distance field and closes training through differentiable rendering [12]. Object-centric occupancy completion instead canonicalizes long LiDAR tracklets and queries an implicit decoder at Actor-relative coordinates [38]. SparseOcc++ separates geometric completion from semantics by regressing an anisotropic scene-completion field on sparse anchors [29]. Gau-Occ initializes semantic Gaussians from a diffusion-completed LiDAR cloud, but its final multimodal head may still update center, scale, and rotation under occupancy supervision [20]. PoinTr formulates completion as set-to-set translation, while SnowflakeNet grows child points from parent seeds [34, 36]. We combine only these target/set principles: dynamic sweeps are aligned per Actor, observed anchors are immutable, image features cannot update geometry, and ray/surface losses remain active. Our target is not occupancy mIoU: it is the joint reduction of literal early returns without degrading held-out Actor surface distance.

GaussianFormer-2 interprets sparse Gaussians as occupied-region probability distributions [13], while GaussRender adds projective depth and semantic losses alongside native 3D occupancy supervision [5]. Vol3DGS analytically integrates 3D Gaussians to obtain volumetrically consistent transmittance [28]. ShelfOcc likewise argues that metrically consistent native-3D targets are essential in dynamic driving scenes [2]. These results motivate our Gaussian energy, but not unconstrained end-to-end geometry: we keep motion-compensated Actor targets and audit whether ray gradients alter centers or merely inflate support.

Feature 3DGS attaches image-distilled descriptors to Gaussian carriers [40], while Neural Shell Texture Splatting explicitly separates surface geometry from a sampled texture field [37]. We use the same narrow separation principle, not their visual objective: image-trained SH and opacity can be associated with a frozen physical carrier, but cannot update its center or scale.

Uncertainty and reliability. Deep ensembles expose useful empirical predictive disagreement [18], and conformalized quantile methods can turn residuals into calibrated marginal intervals under appropriate assumptions [24]. Interval-bound methods can certify a network over a specified input box, including for robust OOD detection [1]; they do not certify that the box models an unseen sensor domain. Driving logs violate many simple exchangeability assumptions through scene correlation and domain shift. We therefore report empirical calibration explicitly, preserve claim boundaries, and use monotone structure only for runtime consistency—not as a formal road-safety guarantee.

Selective prediction couples conditional risk to coverage [8], while learning to defer evaluates the downstream fallback as part of the system [21]. Moreover, abstention can improve aggregate regression yet worsen a predefined subgroup [27]. We therefore report both selected and fallback-composite outcomes and decompose them over the frozen physical hazard stratum; this is descriptive accounting, not subgroup calibration.

Cross-sensor LiDAR generalization. LiDAR domain shift includes sensor sampling as well as scene and appearance shift. Source-only subsampling consistency can improve transfer across scan densities [17], while partially accumulated geometry and sequence cues provide a complementary common representation [25]. These results motivate our frozen sparsity intervention, but do not make source consistency a certificate for actor-level repair selection; we therefore retain exact-once source and external evaluations.

Our distinction. Prior confidence, occupancy, and reconstruction-quality signals often mix physical invalidity, visibility, and behavioral difficulty. HARP-3D instead defines separate evidence inputs, paired interventions, and failure denominators so that an improvement cannot be obtained by deleting hazardous Actors or relabeling unknown space as free.

3. Problem Formulation

Let Actor i have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \tag{1}$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with A_i, but collision queries consume only an Actor-local physical surface S_i^{phys} and its known/unknown mask.

We separate two random variables. Artifactness a_i depends on sensor and geometric evidence E_i, physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \tag{2}$$

Hazardness h_i depends on Actor dynamics, map context, and a candidate Ego trajectory τ:

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \tag{3}$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve a_i. For the same behavior, a clean-to-artifact intervention should preserve h_i.

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \le H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \varepsilon)}. \tag{4}$$

The runtime interface returns $R(\tau, H, b) = P(C \le b)$ and enforces $\partial R / \partial b \ge 0$ and $\partial R / \partial H \le 0$.

4. Method

4.1. Canonical Evidence and Immutable Actor State

For Actor i at time t, annotated sensor, ego, and Actor poses map a LiDAR endpoint into a canonical coordinate system,

$$x_{i,t} = T_{i,t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \tag{5}$$

Build sweeps provide model input; disjoint held-out sweeps construct the target T_i and are never input features. Directly observed and matched projected endpoints form an immutable anchor set A_i. Actor identity, trajectory, extent, lifecycle, and hazard label are immutable interfaces rather than prediction targets.

For a held-out ray $r=(o,v,d^\star)$ and physical surface S, the literal first return is

$$d_S(r)=\min\{s>0: o+sv\in S_\epsilon\}, \qquad \min\varnothing=\infty. \tag{6}$$

An early return satisfies $d_S(r)<d^\star-\epsilon$. Deletion can only make this predicate non-increasing, but may destroy hits and surface coverage. We therefore do not treat a lower early-return rate as sufficient evidence of consistency.

4.2. Supervision-Native Child Surfaces

One completion seed cannot cover a dense target surface. Each seed x_j predicts four bounded Actor-frame residuals and positive support scales,

$$x'_{jk}=x_j+\Delta x_{jk}(E_i), \qquad S_i^\theta=A_i\cup\{(x'_{jk},s_{jk})\}_{j,k=1}^4, \tag{7}$$

where E_i contains build-only geometric and ray evidence. Geometry pretraining combines bidirectional set distance, target-local point-to-plane distance, and a target-spacing scale loss. Physical fine-tuning keeps these terms active and adds frame-balanced target coverage, differentiable literal first return, and observed free-before-hit supervision:

$$\mathcal{L}_{\text{phys}}=\mathcal{L}_{\text{set}}+.25\mathcal{L}_{\text{plane}}+.1\mathcal{L}_{\text{scale}}+\mathcal{L}_{\text{frame}}+\mathcal{L}_{\text{first}}+.5\mathcal{L}_{\text{free}}. \tag{8}$$

Space behind the first hit is not labeled free. Every child remains in S_i^θ at deployment; UNKNOWN is evidence state, not a mask. Thus an improvement must arise through the supervised representation rather than selective output removal.

4.3. Producer-Evidential Return Authority

Surface support and sensor termination are distinct. For each anchor or child j, the build descriptor e_j retains source ray/frame identity, construction provenance, projection displacement, canonical hit count, temporal/view support, and continuous build evidence $m_j^{\text{build}}=(m_F,m_O,m_U)$. Disjoint target rays construct m_j^{GT} from free evidence before a return, occupied evidence near it, and unknown evidence behind it or outside ray support. A bounded head predicts $m_j=f_\phi(e_j)$ and is supervised by soft three-state cross-entropy. Completion children also receive the GT free-before-hit event loss used by the deployed ray objective. Geometry is frozen during this authority stage.

Let o_j be predicted occupied mass and κ_{kj} the frozen Gaussian support of primitive j at ordered depth d_k. Instead of adding $o_j\kappa_{kj}$ as density-dependent opacity, we define one joint categorical return measure,

$$q(k,j)=\frac{o_j\kappa_{kj}}{\sum_{a,b}o_b\kappa_{ab}}, \qquad p_k=\sum_j q(k,j), \quad F_k=\sum_{\ell\le k}p_\ell. \tag{9}$$

The deployed return is the median, $\widehat{d}=\min\{d_k: F_k\ge 1/2\}$. Training and deployment use the same ordered distribution. Normalization guarantees non-negative unit mass and a monotone CDF, but does not turn o_j into occupancy or collision probability. The current protocol conditions on one return inside the Actor box and does not model ray drop.

For boundary $b=d^\star-\epsilon$, piecewise-linear interpolation gives the exact deployed decision

$$\widehat{d}<b \quad\Longleftrightarrow\quad F(b)>\tfrac{1}{2}, \qquad F(b)=F_{\text{anchor}}(b)+F_{\text{child}}(b). \tag{10}$$

This decomposition explains responsibility but is never used to delete a primitive or case.

4.4. Typed Motion and Appearance Ownership

Rigid pose is a separate read-only authority. With $T_{it}=(R_{it},t_{it})$, the physical query is

$$Q_{it}(x;P_i,T_{it})=\log\sum_j\exp\left[-\frac{\|R_{it}^\top(x-t_{it})-c_{ij}\|_2^2}{2s_{ij}^2}\right], \tag{11}$$

where $P_i=\{(c_{ij},s_{ij})\}_j$ is canonical physical state. Static Background is a separate scene component. Visual state V_i contains render geometry, SH, and opacity; image optimization receives $\mathrm{sg}(P_i)$ and updates only V_i. Consequently $\partial Q_{it}/\partial V_i=0$, and for a common rigid transform g, inverse querying yields $Q_{gT}(gx)=Q_T(x)$. These are ownership and equivariance guarantees, not guarantees that pose or geometry is correct.

4.5. Finite Attenuation Boundary

The normalized measure also identifies when learned visibility is unsafe. For primitive j, let $D_j=\sum_k\kappa_{kj}$, $N_j=\sum_{d_k<b}\kappa_{kj}$ with boundary interpolation, $C_j=N_j/D_j$, and $r_j=o_jD_j/\sum_a o_aD_a$. Direct differentiation gives

$$\frac{\partial F(b)}{\partial\log o_j}=r_j\big(C_j-F(b)\big). \tag{12}$$

For any finite attenuation $v\in(0,1]$ applied to the component, or to a family treated as one component,

$$F_v(b)-F(b)=\frac{(1-v)r_j\big(F(b)-C_j\big)}{1-(1-v)r_j}. \tag{13}$$

Attenuation lowers the pre-boundary CDF iff $C_j>F(b)$; changing its magnitude cannot reverse the sign. Because b uses target depth, this is an explanation and model-design boundary, not a deployable oracle gate.

A distinct non-implication concerns the geometry surrogate. The differentiable renderer optimizes an alpha-composited expected depth, whereas Eq. 6 is a minimum over beam-supported surface. An arbitrarily small shallow

Surface	Free (%) ↓	Recall (%) ↑	CD (m) ↓
Single-frame input	100.0	53.36	0.254
Nearest canonical	84.91	**62.60**	**0.168**
Ray-certified	**0.00**	59.56	0.175

Table 1. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

component can move the minimum across b while perturbing the expectation arbitrarily little; with late residual mass, that perturbation may even move the expectation toward the target. Lower expected-depth error therefore cannot certify a safer literal first return without an explicit ordered support or CDF margin.

5. Experiments

5.1. Protocol and Claim Boundary

We train and make every representation choice on nuScenes. The geometry fold contains 66 Actors and is explicitly marked as development-exposed because its parent representation had already been inspected. Argoverse 2 Sensor validation is a single frozen zero-shot evaluation: before any learned-method output is read, we sort the 150 log UUIDs and choose every fifth log, yielding 20 quantitative logs. We prohibit AV2 fine-tuning, calibration, threshold selection, and failed-log deletion. At the time of this draft the aggregate is still incomplete, so no partial learned-transfer metric appears below.

Physical evaluation uses literal first-return error, hit recall within $0.20\,\mathrm{m}$, symmetric Chamfer distance, and the same metrics stratified by a target-defined hazard label. Actor identity, trajectory, extent, and lifecycle are immutable. This prevents a method from improving an early-return score by removing a difficult Actor. Unless stated otherwise, all learned comparisons keep every Actor and every generated surface.

As an external, non-learned reference, the frozen ray-certified compiler operates on 433 AV2 Actors, 147 hazardous (Table 1). It reduces target depth error from 0.207 to $0.154\,\mathrm{m}$ and Chamfer from 0.254 to $0.175\,\mathrm{m}$ while preserving every Actor. Nearest-surface projection obtains lower Chamfer but violates matched observed-free rays, illustrating why average geometry alone is not a physical objective.

5.2. Supervision-Native Canonical Geometry

The train-only displacement oracle shows that a non-deletion Pareto direction exists, but a residual policy reproduces its ray gain mainly through an UNKNOWN action. We therefore remove the action and change the target. Motion-corrected held-out LiDAR is transformed into each Actor's canonical frame; every completion seed predicts four child centers and scales, and training keeps target-set, tangent-plane, scale, per-frame coverage, literal first-return, and free-before-hit losses active. No output is filtered at deployment.

Table 2. Core source-domain results on the frozen, exposed nuScenes development fold. Negative early-return and Chamfer changes are favorable; positive hit changes are favorable. Comparisons differ by panel and are stated explicitly.

Stage	Metric	Change
Geometry: M8 vs. original completion		
	hazardous early returns (relative)	-5.123%
	symmetric Chamfer	$-6.207\,\mathrm{mm}$
	hit recall	$+2.759\,\mathrm{pp}$
Return measure: M39 vs. unit categorical energy		
	early returns (all / hazard / clear)	$-0.52/-0.62/-0.03\,\mathrm{pp}$
	hit recall (all / hazard / clear)	$+2.17/+2.32/+1.46\,\mathrm{pp}$

Table 2 reports the resulting M8 model. Hazardous early returns decrease by 5.12% relative, Chamfer improves by 6.21 mm, and hit recall rises by 2.76 points. The three metrics move together because physical consistency is imposed on the target and loss, rather than obtained by deleting low-confidence surfaces. The representative Actor in Fig. 1 is selected by the *baseline* early-return median before inspecting model output.

5.3. Producer Evidence as a Categorical Return Measure

Geometry and ray termination are not the same variable. Exact producer replay over 1,004 Actors aligns 297,535 anchors with build and held-out FREE/OCCUPIED/UNKNOWN votes. Their occupied masses correlate ($r = 0.514$), but direct three-state agreement is only 43.41%; 58.72% of anchors receive both free and occupied held-out votes. Thus provenance is informative but cannot be a hard keep/delete label.

We freeze M8 geometry and learn only continuous authority from build-time producer evidence. Weighting Gaussian energy or interpreting authority as additive optical thickness fails: opacity grows with primitive density and couples endpoint termination to the front tail. The decisive M39 change leaves the learned masses untouched and interprets them as a normalized categorical surface-return measure. Relative to unit categorical energy, early returns decrease in all, hazardous, and clear strata by $0.52/0.62/0.03$ points, while hit recall rises by $2.17/2.32/1.46$ points (Table 2). These three pre-registered directions pass without changing geometry or removing a primitive. The conclusion is deliberately narrow: the mass is return evidence, not occupancy probability or collision risk.

Failure-directed variants sharpen that conclusion. Joint

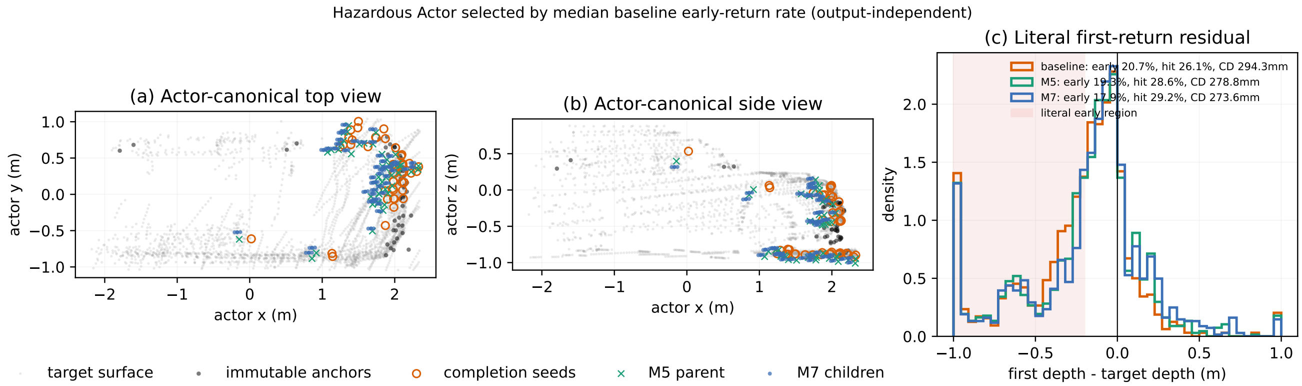

Figure 1. Supervision-native geometry on a hazardous development Actor selected without consulting the learned output. Forty-seven completion seeds expand into 188 child surfaces while observed anchors remain fixed. Baseline/M5/M7 early rates are 20.7/19.3/17.9%, hit rates are 26.1/28.6/29.2%, and Chamfer distances are 294.3/278.8/273.6 mm. Gray held-out GT points are target-only.

ray fine-tuning transfers mass to the denser child family; fixing total family mass still optimizes target-bin likelihood without controlling the deployed boundary event. Directly supervising GT not-early and hit-band events recovers aggregate and hazardous directions but regresses the clear stratum. We therefore retain frozen M39 for the single AV2 test instead of tuning another development loss.

5.4. An Exact Boundary for Learned Attenuation

M44 verifies the exact CDF accounting of Eq. 10 over 99,208 rays. M39 lowers mean pre-boundary mass by 0.331 points relative to unit energy: observed anchors contribute -0.346 points, while completion children contribute $+0.016$ points. The supported gain is therefore not a completion filter; producer-supervised anchor authority dominates it.

Local oriented support and a bounded learned visibility factor both lower their training losses but transport error between ray strata. The visibility model raises hazardous early returns by 0.483 points relative to M39. Equation 13 explains why tuning its magnitude cannot repair the sign. In Fig. 2, uniform child attenuation is analytically adverse on 58.85% of rays and safe on only 30.96%. Under the learned component-wise attenuation, adverse pressure dominates 65.45% overall and the exact pre-boundary CDF rises on 59.32% of rays. The first-order sign agrees with the finite change on 95.73% of rays. We use this identity only to explain the failure—never as an oracle gate or post-hoc selector.

M51 tests the separate expected-depth/minimum boundary using M8 and rejected M50. On identical, non-voxelized support, hard early rate rises by 0.490 points; 38.03% of the newly early rays nevertheless have lower smooth absolute depth error. Under the full-anchor, voxelized deployment surface, the values are 0.545 points and

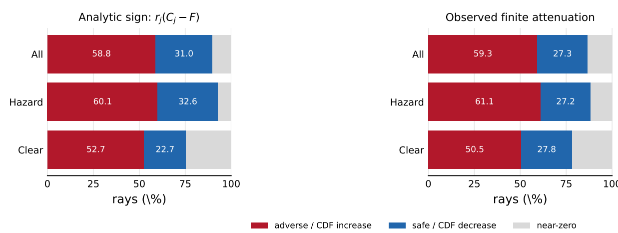

Figure 2. Exact attenuation boundary on 99,208 frozen rays. Red is analytically adverse (left) or an observed increase in pre-boundary CDF (right); blue is safe/decreasing. The diagnostic is not a deployment rule.

36.54%. Smooth-error and hard-depth changes have correlation 0.038, while support realization flips 1.28% of ray events. The non-implication therefore precedes deployment discretization and is not repaired by better frame weighting.

5.5. Typed Motion, Static Context, and Appearance

We audit the ownership contract separately from surface accuracy. Twelve scene-0230 Actors are identity-matched to a read-only StreetGS checkpoint. Across 36 Actor–frames, 5,791 physical Gaussians follow the checkpoint's rigid Actor poses, while 106,807 appearance-owned and 1,140,862 Background Gaussians remain outside the physical query. Inverse querying gives maximum energy and pairwise-distance residuals of 3.40×10^{-14} and 1.24×10^{-14} m under as much as 126.28 m translation. This establishes the SE(3) composition contract, not pose or geometry correctness.

Image supervision is likewise prevented from moving collision geometry. Optimizing only SH and opacity on a fixed 309-carrier Actor improves held-out footprint PSNR from 15.86 to 16.94 dB, but remains 8.41 dB below the original visual Actor. A fixed parent–surfel hierarchy improves all six held-out views to 17.73 dB, yet retains a 7.62 dB gap.

The result supports typed gradient ownership and exposes insufficient appearance capacity; it does not justify sharing image-trained geometry with the physical field.

5.6. What the Evidence Establishes

The source experiments support three separable statements: GT-defined losses improve Actor-canonical geometry without deletion; producer evidence is useful when represented as categorical return authority; and normalized-mixture attenuation has a non-monotone, exactly characterizable boundary. Typed ownership keeps rigid motion, Background, appearance, and canonical geometry from silently compensating for one another. None of these statements certifies collision freedom, unobserved completion, trajectory accuracy, domain invariance, or real-road safety. The frozen AV2 aggregate will test only the learned return representation's cross-domain direction; its outcome cannot retroactively select or adapt the method.

6. Limitations

The learned results use a 66-Actor development fold whose parent representation was previously inspected. They establish a source-domain Pareto direction, not an unbiased final estimate. The single frozen 20-log AV2 aggregate is still pending; we neither inspect partial metrics nor adapt on AV2. The earlier non-learned AV2 compiler is a separate operator result and must not be presented as learned transfer.

Our categorical measure assumes one return inside an annotated Actor box. Its mass is surface-return evidence, not calibrated occupancy or collision probability; sparse sensing leaves unknown space, and neither M8 nor M39 proves watertight completion. Tracked cuboids, poses, and GT LiDAR may be noisy. Current canonicalization covers rigid annotated vehicles, while deformable Actors, static topology, explicit no-return modeling, sensor-error sets, and closed-loop vehicle dynamics remain open. Exact SE(3) equivariance certifies coordinate composition only, not trajectory or surface correctness.

The appearance audit is a single-scene mechanism study on views exposed before the final hierarchy design. It shows that image gradients can be isolated from physical geometry, but leaves a 7.62 dB gap to the original visual Actor and does not establish photorealism or cross-scene fidelity. Finally, the M49 attenuation identity is evaluated with a GT-defined boundary on exposed rays. It explains why attenuation can be adverse, but using its sign as a selector would recreate the post-hoc oracle behavior we seek to avoid. None of our results is a collision-freedom, planning, policy, domain-invariance, or real-road-safety guarantee.

7. Conclusion

Physical consistency is a property of targets, supervision, representation, and deployment together—not a score that can be repaired by deleting uncertain outputs. Motion-corrected Actor targets and persistent set/ray/free-space losses let our child-surface model improve early returns, hit recall, and Chamfer jointly on nuScenes while retaining every Actor. Producer evidence then becomes useful only when interpreted as categorical surface-return authority, not additive volumetric opacity. Our negative ablations sharpen this distinction: local orientation and learned visibility transport risk between ray strata even as their training losses fall. The exact finite attenuation boundary explains why no opacity magnitude can make this operation universally safe. Typed ownership and SE(3) inverse querying keep canonical geometry, rigid motion, static context, and appearance separate, but do not certify their accuracy. The pending frozen AV2 aggregate will determine whether the learned return representation transfers; regardless of that outcome, the source-domain supervision result and normalized-mixture safety boundary do not imply collision freedom, domain invariance, or real-road safety.

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